

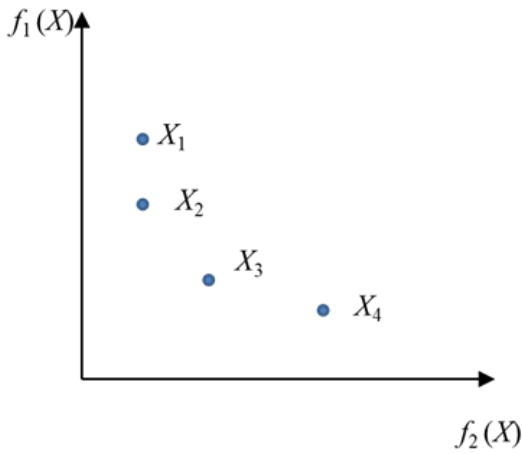
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Next item →

1. The Pareto front of the multi-objective optimization problem $\min[f_1(X), f_2(X)]$ is shown in the following figure. Which of the following statement about dominance is correct?

1 / 1 point



- ☐ X1 dominates X4
- ☒ X2 dominates X1
- ☐ X2 dominates X4
- ☐ X1 dominates X2 and X3

👉 Nice work

2. What is a combinatorial optimization problem?

1 / 1 point

- ☐ A combinatorial optimization problem refers to a multi-objective optimization problem where the decision variables can only take discrete values, such as integers or elements from a discrete set.
- ☒ A combinatorial optimization problem refers to an optimization problem where the decision variables can only take discrete values, such as integers or elements from a discrete set.
- ☐ A combinatorial optimization problem refers to an optimization problem where there is no restriction on the search ranges of the decision variables.
- ☐ A combinatorial optimization problem refers to an optimization problem where the decision variables can take any real values.

👉 Nice work

3. What is the global optimum in a single objective unconstrained optimization problem?

1 / 1 point

- ☒ The global optimum refers to the solution where the objective function reaches its minimum or maximum value in the entire search space.
- ☐ The global optimum refers to the solution with an objective function value of 0.
- ☐ The global optimum refers to a solution where the objective function reaches its minimum or maximum value in a neighborhood.

👉 Nice work

4. Given an optimization problem which is formulated as
$$\begin{cases} \min f(x) \\ \text{s.t. } g(X) \leq 0 \\ h(X) = 0 \\ X = [x_1, x_2, \dots, x_n]^T \end{cases},$$

1 / 1 point

what is the decision variable of the optimization problem in the above equation?

- ☐ $f(X)$
- ☒ $X = [x_1, x_2, \dots, x_n]^T$
- ☐ $h(X) = 0$
- ☐ $g(X) \leq 0$

👉 Nice work

5. Which of the following optimization problem is a multimodal problem?

1 / 1 point

- ☐ $f(x) = x^2$
- ☒ $f(x) = 1 - \sin(2x)$
- ☐ $f(x) = x - 2$
- ☐ $f(x) = 3 - 0.1x^3$

👉 Nice work